

SIMATS ENGINEERING

ASSESSMENT TOOL 1: Class Test

COURSE NAME: Cloud Computing and
Big Data Analytics

COURSE CODE: CSA1522

COURSE OUTCOME COVERED

CO4: *Analyze big data characteristics and challenges across domains including security and application aspects.*
(BL4)

Dave's Taxonomy: Articulation (Level 4)
Question Count: 10

Total Marks: 30

Code of Conduct

I Bala S (192524427) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: bala

Assessment Tasks

Class Test

1. Analyze a multinational retail organization struggling with fragmented customer data across online and offline channels, and propose a unified big data architecture that enables real-time customer analytics and personalized recommendations.
2. Evaluate the effectiveness of distributed computing models in processing large-scale scientific research data, and justify how parallel processing improves computational efficiency and scalability compared to centralized systems.
3. Design a fraud detection framework for a digital payment platform using big data analytics, and justify how streaming data processing and predictive models improve fraud prevention accuracy.
4. Assess the challenges of storing and processing multimedia data such as video, audio, and images in a big data environment, and recommend suitable technologies and storage mechanisms for efficient management.
5. Construct a disaster recovery and business continuity strategy for a cloud-based big data platform, and justify how redundancy, replication, and automated failover mechanisms ensure operational resilience.
6. Analyze the impact of data quality issues such as inconsistency, duplication, and missing values in a large-scale analytics platform, and propose data governance strategies to improve analytical reliability.
7. Design a scalable recommendation system for an online streaming service using big data technologies, and justify how user behavior analytics and distributed processing enhance recommendation accuracy and user engagement.
8. Evaluate the role of cloud computing in modern big data ecosystems, and critically examine its advantages and limitations in terms of scalability, cost management, flexibility, and security.
9. Develop a real-time analytics framework for transportation and logistics management, and justify how big data technologies improve route optimization, fuel efficiency, and operational decision-making.

10. Critically analyze ethical and legal challenges associated with big data collection and analytics, and recommend organizational policies that ensure responsible data usage, regulatory compliance, and user trust.

Class Test Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Concept Accuracy	30%	Accurate and well-expressed technical answers	Mostly accurate answers	Partial accuracy	Inaccurate answers
Coverage	20%	Covers all required points	Covers most points	Limited coverage	Very poor coverage
Use of Examples	15%	Uses relevant and meaningful examples	Some relevant examples	Weak examples	No useful examples
Analytical Awareness	20%	Shows strong awareness of issues and implications	Reasonable awareness	Basic awareness	Minimal awareness
Clarity	15%	Clear and concise writing	Generally clear	Some unclear expressions	Poor clarity

1. Unified Big Data Architecture for Retail

- A multinational retail organization may have customer data stored separately in websites, mobile apps, physical stores, CRM systems, and social media.
- A unified big data architecture can use Kafka for real-time data ingestion, HDFS ,Spark for distributed processing, and NoSQL databases for fast customer-profile access.
- Real-time streams can be processed using Spark Streaming or similar technologies to identify customer behavior immediately.

Benefits:

- 360° customer view
- Real-time customer analytics
- Personalized recommendations
- Improved customer experience

2. Distributed Computing for Scientific Research

- Scientific research produces massive datasets from simulations, satellites, sensors, genomics, and experiments.
- Distributed computing divides a large task into smaller tasks and executes them simultaneously across multiple machines.
- For example, a scientific dataset containing billions of records can be divided among several computing nodes.

Parallel processing improves efficiency because:

- Multiple operations execute simultaneously.
- Processing time is reduced.
- Workloads are distributed across machines.
- Additional nodes can be added when data increases.

3. Big Data Fraud Detection Framework

- A digital payment platform can design a fraud detection system that continuously monitors transactions.
- Transaction information such as amount, location, device, time, frequency, and customer history is collected in real time.
- Kafka can ingest high-volume transaction streams, while Spark Streaming or another stream-processing system analyzes them.
- For example, a transaction may be considered suspicious when it differs significantly from the customer's normal behavior.

Advantages:

- Real-time fraud detection
- Faster response to suspicious transactions
- Detection of previously unknown patterns
- Reduced false positives using predictive models
- Continuous model improvement using historical data

4. Multimedia Data in Big Data

- Multimedia data includes images, audio, and video. It creates challenges because of its large size, complex formats, high processing requirements, and unstructured nature.
- Traditional relational databases are not ideal for storing huge multimedia files.
- A big data system can use metadata databases or NoSQL databases to store information such as file name, type, location, timestamp, and user ID.
- Distributed processing frameworks such as Spark can process large multimedia datasets.

Suitable technologies:

- Object storage – large multimedia files
- NoSQL – metadata management
- Hadoop/Spark – distributed processing
- CDN – fast content delivery

5. Disaster Recovery and Business Continuity

- A cloud-based big data platform must continue operating even when hardware, software, or network failures occur.
- A suitable disaster recovery strategy should include redundancy, replication, backups, monitoring, and automated failover.
- Data can be replicated across multiple availability zones or regions. Critical databases and storage systems should maintain multiple copies.
- Automated monitoring can detect failures and redirect workloads to healthy resources.

Key mechanisms:

1. **Redundancy:** Multiple servers prevent a single point of failure.
2. **Replication:** Multiple copies of data protect against data loss.
3. **Backup:** Periodic backups allow recovery from major failures.
4. **Automated failover:** Traffic is automatically redirected to another system.
5. **Monitoring:** Continuous monitoring detects failures quickly

6. Data Quality and Governance

Major problems include:

- **Inconsistency:** Different systems contain different values.
- **Duplication:** The same customer or transaction appears multiple times.
- **Missing values:** Important fields are unavailable.

- **Incorrect data:** Wrong or outdated information produces misleading results.

Governance strategies:

- Data validation rules
- Master Data Management
- Deduplication
- Data quality monitoring
- Metadata management
- Data ownership and stewardship
- Regular auditing

7. Big Data Recommendation System

- An online streaming service can develop a scalable recommendation system using user behavior data such as watch history, searches, ratings, likes, viewing duration, and skipped content.
- Data can be collected using Kafka, stored in a data lake, and processed using Spark.
- Machine learning algorithms can identify similarities between users and content.
- The system can use collaborative filtering, content-based filtering, or hybrid recommendation techniques.

Benefits:

- Personalized recommendations
- Better prediction of user preferences
- Real-time recommendation updates
- Scalable processing

8. Role of Cloud Computing in Big Data

Advantages:

- **Scalability:** Resources can be increased or decreased according to workload.
- **Cost management:** Organizations can use pay-as-you-go services.
- **Flexibility:** Different storage, databases, analytics, and AI services can be integrated.
- **Availability:** Cloud providers offer highly available infrastructure.
- **Faster deployment:** Big data environments can be created quickly.

Limitations:

- Data security and privacy risks
- Dependence on internet connectivity
- Vendor lock-in
- Unexpected cloud costs
- Compliance challenges
- Data migration difficulties

9. Real-Time Transportation and Logistics Analytics

- Real-time data processing can identify traffic congestion, vehicle location, fuel consumption, and delivery delays.
- Machine learning algorithms can recommend alternative routes based on traffic and historical patterns.
- Vehicle sensor data can also identify inefficient fuel usage and predict maintenance requirements.

Benefits:

- Optimized delivery routes
- Reduced travel time
- Improved fuel efficiency
- Reduced vehicle downtime
- Accurate delivery estimates
- Faster operational decisions

10. Ethical and Legal Challenges of Big Data

Major challenges:

- Privacy violations
- Unauthorized data collection
- Lack of user consent
- Data breaches
- Algorithmic bias
- Discrimination
- Excessive data retention
- Lack of transparency